American River College – Math 372
Spring 2020 – Class 23/32

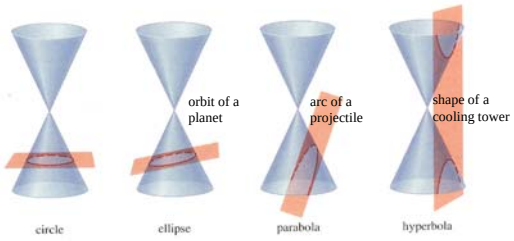
Instructor:	John Burke
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Telephone:	916-484-8601
Class:	A & S 102, TTh 8:15 – 10:20
Student hours:	Howard Hall 148, MW 1:00 – 2:30, TTh 1:00 – 2:00

Today:	§§ 11.1
Assignment:	Online Homework #11
Next:	§§ 11.2; Homework #5 Due; Catch-up/Review

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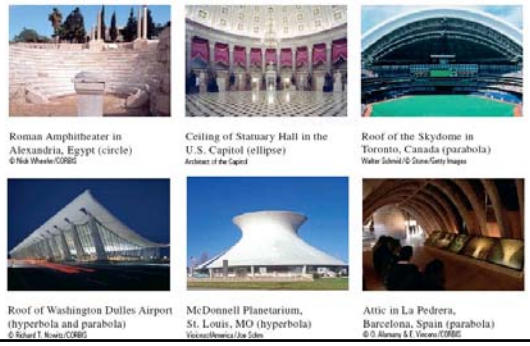
11 Conic Sections

Conic sections are the curves formed by the intersection of a plane with a pair of right circular cones. These curves have 4 basic shapes: **circles**, **ellipses**, **parabolas**, and **hyperbolas**.



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11 Conic Sections

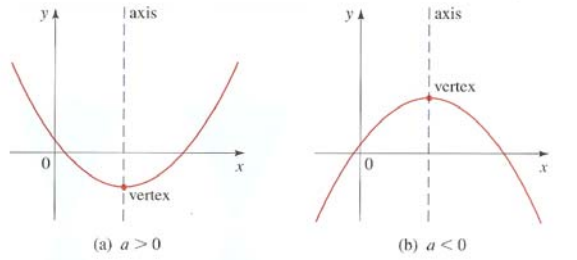


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11.1 Parabolas

Algebraic Definition: $y = ax^2 + bx + c$

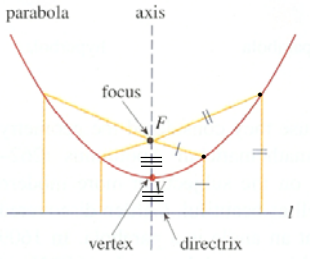
Vertex: $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$



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11.1 Parabolas

Geometric definition: A **parabola** is the set of points in the plane equidistant from a fixed-point ***F*** called the **focus** and a fixed line ***l*** called the **directrix**.



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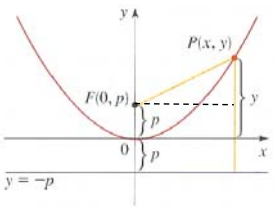
11.1 Parabolas

For now we restrict our discussion to parabolas with the **vertex at the origin** and with **a vertical or horizontal axis of symmetry**. If the focus is at $(0, p)$, then the axis is vertical and the directrix has equation $y = -p$.

The distance from F to P is $\sqrt{x^2 + (y - p)^2}$.

The distance from P to the directrix is $|y + p|$.

Setting them equal yields …



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11.1 Parabolas

PARABOLA WITH VERTICAL AXIS

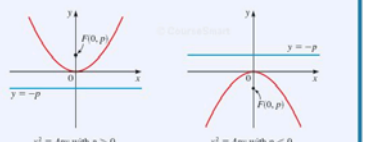
The graph of the equation

$$x^2 = 4py$$

is a parabola with the following properties.

VERTEX	$V(0, 0)$
FOCUS	$F(0, p)$
DIRECTRIX	$y = -p$

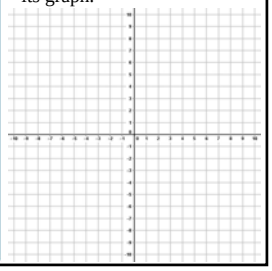
The parabola opens upward if $p > 0$ or downward if $p < 0$.



$x^2 = 4py$ with $p > 0$

$x^2 = 4py$ with $p < 0$

(a) Find the equation of the parabola with vertex $V(0, 0)$ and focus $F(0, 2)$, and sketch its graph.



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11.1 Parabolas

PARABOLA WITH VERTICAL AXIS

The graph of the equation

$$x^2 = 4py$$

is a parabola with the following properties.

VERTEX	$V(0, 0)$
FOCUS	$F(0, p)$
DIRECTRIX	$y = -p$

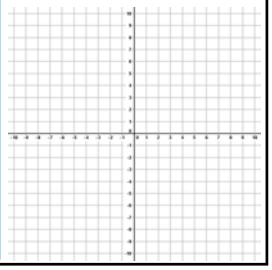
The parabola opens upward if $p > 0$ or downward if $p < 0$.



$x^2 = 4py$ with $p > 0$

$x^2 = 4py$ with $p < 0$

(b) Find the focus and directrix of the parabola $y = -x^2$ and sketch its graph.



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11.1 Parabolas

PARABOLA WITH HORIZONTAL AXIS

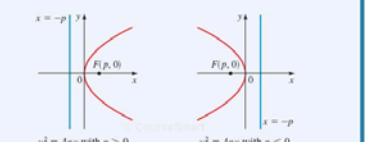
The graph of the equation

$$y^2 = 4px$$

is a parabola with the following properties.

VERTEX	$V(0, 0)$
FOCUS	$F(p, 0)$
DIRECTRIX	$x = -p$

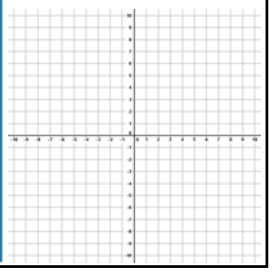
The parabola opens to the right if $p > 0$ or to the left if $p < 0$.



$y^2 = 4px$ with $p > 0$

$y^2 = 4px$ with $p < 0$

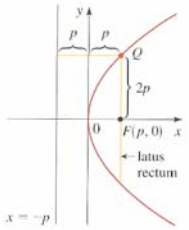
Find the focus and directrix of the parabola $6x + y^2 = 0$.



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11.1 Parabolas

The line segment that runs through the focus perpendicular to the axis, with endpoints on the parabola, is called the **latus rectum**, and its length is called the **focal diameter** of the parabola.

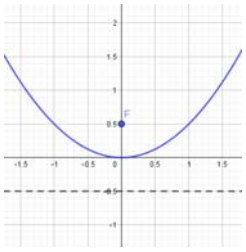


Note: focal diameter = 4*p* from the definition.

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11.1 Parabolas

The line segment that runs through the focus perpendicular to the axis, with endpoints on the parabola, is called the **latus rectum**, and its length is the **focal diameter** of the parabola.



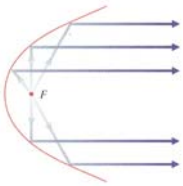
Note: focal diameter = 4*p* from the definition.

Find the focus, directrix, and focal diameter of the parabola $y = \frac{1}{2} x^2$ and sketch the graph.

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11.1 Parabolas

Light from a source placed at the focus of a surface with parabolic cross-section will be reflected parallel to the axis of symmetry. Think search lights.

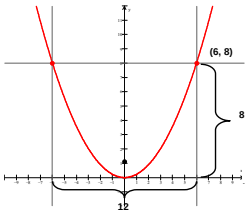


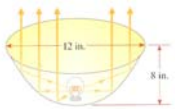
Conversely, light approaching the reflector in rays parallel to the axis of symmetry is concentrated to the focus. This is used in large reflecting telescopes (also parabolic mikes where a microphone is placed at the focus).

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11.1 Parabolas

Example: A searchlight has a parabolic reflector that forms a bowl, which is 12" wide from rim to rim and 8 in. deep. If the filament of the light bulb is located at the focus, how far from the vertex of the reflector is it?





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11.1 Parabolas

Your Turn:

1. Let $x^2 = 5y$ be the equation of a parabola. Find the focus, directrix, and focal diameter of the parabola. Sketch the graph of the parabola and the directrix.
2. Find an equation for the parabola that has its vertex at the origin and satisfies the condition that the focus is at $(0, -\frac{3}{4})$.
3. Find an equation for the parabola that has its vertex at the origin and satisfies the condition that the directrix is $x = -4$.
4. Find an equation for the parabola that has its vertex at the origin and satisfies the condition that it opens downward with focus 10 units away from the vertex.

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